

ME 333
MECHANICS OF MATERIALS

Lecture 13: Yield Criterion &
Tutorial Problems

Ravi Sastri Ayyagari

Mechanical Engineering
Indian Institute of Technology Gandhinagar

Sem I, 2025-2026

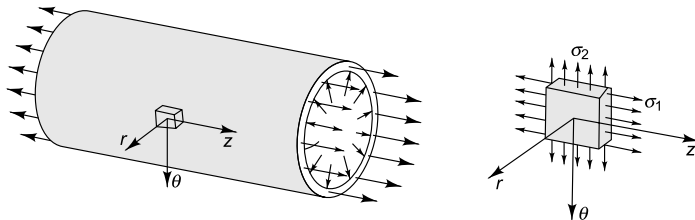
Yield Criterion
Criterion for Initial Yielding

Q What is the need for a yield criterion - if a simple uniaxial test can give you the yield strength?

Yield Criterion
Criterion for Initial Yielding

Q What happens in case of a generalized loading scenario?

Thin-walled Cylinders :

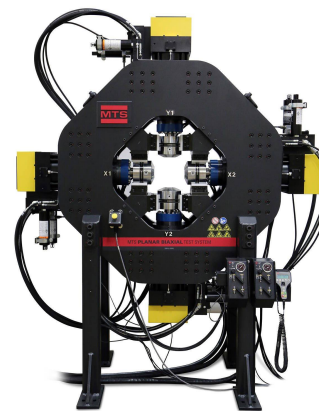


- Yielding is unaffected by hydrostatic state of stress
- Maximum shear stress is half the difference of the principal stresses (*Mohr's circle*)

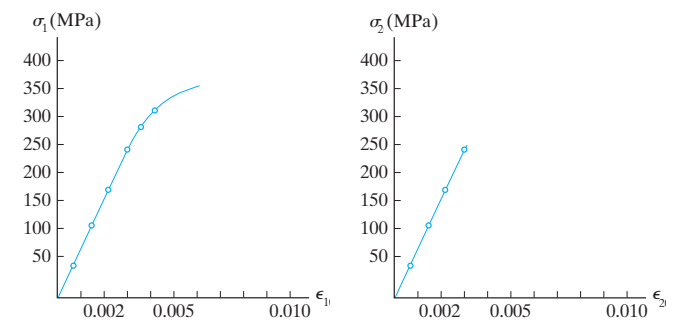
Ref: An Introduction to Mechanics of Solids, Crandall

Yield Criterion
Criterion for Initial Yielding

Biaxial Loading Scenario :



Ref: MTS



Has the material yielded or not ?

Yield Criterion

- A **yield criterion** is used to predict if a material yields under a given state of stress or strain that corresponds to a state experienced by the material under true operating conditions.
- In general, a yield criterion is usually defined by means of a yield function

$$f(\sigma, Y) = 0$$

- Based on the inequalities:

$$\begin{aligned} f(\sigma, Y) &< 0 \quad (\text{elastic}) \\ &= 0 \quad (\text{onset of yielding}) \\ &> 0 \quad (\text{undefined}) \end{aligned}$$

- Typically, $f(\sigma)$ is known as effective stress measure σ_{eff}

Maximum Principal Stress Criterion

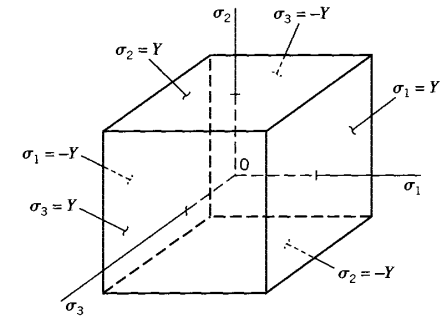
Rankine Criterion

- Yielding occurs when the maximum principal stress equals uniaxial yield stress.

$$f = \max(|\sigma_1|, |\sigma_2|, |\sigma_3|) - Y$$

- Corresponding yield surface is defined by:

$$\sigma_1 = \pm Y; \quad \sigma_2 = \pm Y; \quad \sigma_3 = \pm Y$$

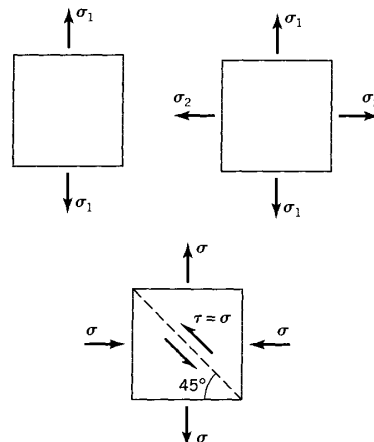


Maximum Principal Stress Criterion

Rankine Criterion

Drawbacks:

- Ignores the effect of other principal stresses
- Only valid if shear stress of material is equal to the uniaxial tensile yield stress



NOTE: Good for predicting the failure under tension in brittle materials.

Maximum Principal Strain Criterion

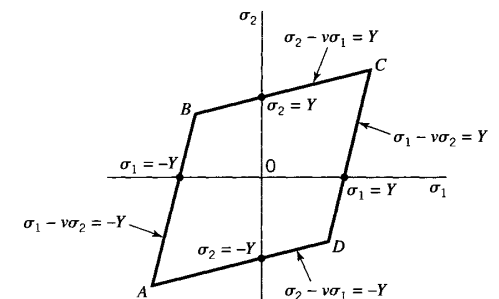
St. Venant's Criterion

- Yielding occurs when the maximum principal strain equals uniaxial yield strain $\epsilon_Y = Y/E$.

$$f = \max(|\epsilon_1|, |\epsilon_2|, |\epsilon_3|) - (Y/E)$$

- Corresponding yield surface is defined by:

$$\begin{aligned} \epsilon_1 &= \frac{1}{E} (\sigma_1 - \nu\sigma_2 - \nu\sigma_3) = \pm \frac{Y}{E} \\ \Rightarrow \sigma_1 - \nu\sigma_2 - \nu\sigma_3 &= \pm Y \\ \sigma_2 - \nu\sigma_3 - \nu\sigma_1 &= \pm Y \\ \sigma_3 - \nu\sigma_1 - \nu\sigma_2 &= \pm Y \end{aligned}$$

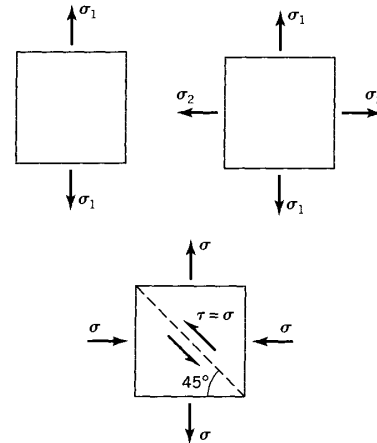


Maximum Principal Strain Criterion

St. Venant's Criterion

Drawbacks:

- If σ_2 is positive then $\sigma_1 > Y$ results in yielding.
- If σ_2 is negative then $\sigma_1 < Y$ results in material to yield.
- Does not predict the yielding in ductile materials accurately.



NOTE: Good for improving the failure predictions in brittle materials.

Ref: Advanced Mechanics of Materials, Boreis & Schmidt

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Strain energy density Criterion

Beltrami Criterion

- Yielding occurs when the strain energy density equals that in yielding under uniaxial tension.

$$U_0 = U_{0Y} = \frac{Y^2}{2E}$$

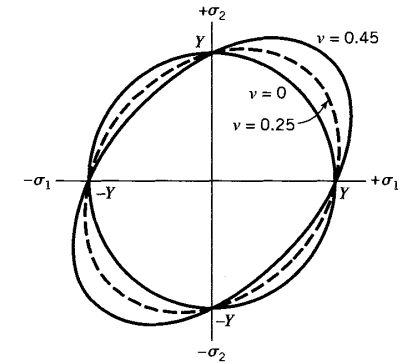
where:

$$U_0 = \frac{1}{2E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)]$$

$$= \frac{\sigma_e^2}{2E}$$

- Corresponding yield surface is defined by:

$$f = \sigma_e^2 - Y^2 = 0$$



Ref: Advanced Mechanics of Materials, Boreis & Schmidt

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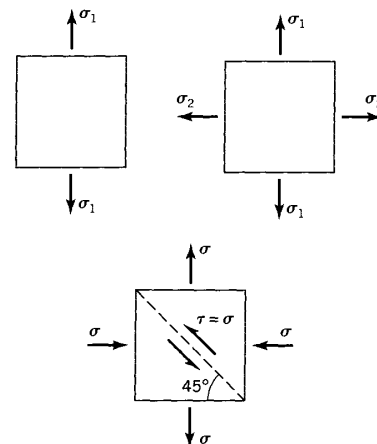
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Strain energy density Criterion

Beltrami Criterion

Drawbacks:

- For uniaxial tension the yielding prediction results in $\sigma_1 = Y$.
- In case of biaxial loading (equi-biaxial) yielding occurs when $2\sigma^2(1 - \nu) = Y^2$ and if the Poisson's ratio is zero results in $\sigma = \sigma_1 = \sigma_2 = Y/\sqrt{2}$.
- Even under shear loading the yield prediction results in $\sigma = Y/\sqrt{2}$ when Poisson ratio is zero.



NOTE: In a full three-dimensional setting the yield surface is an ellipsoid.

Ref: Advanced Mechanics of Materials, Boreis & Schmidt

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Maximum Shear Stress Criterion

Tresca Criterion

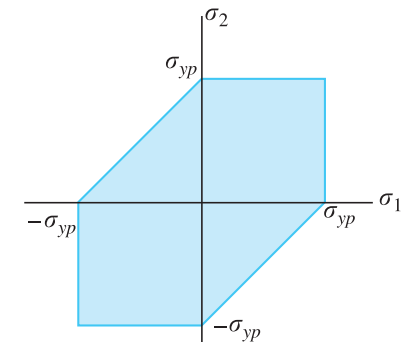
- Yielding occurs when the maximum shear stress equals maximum shear stress at yield under uniaxial tension.

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \tau_Y = \frac{Y}{2}$$

- Corresponding yield surface is defined in respective planes.

$$\tau_{\max} = \frac{|\sigma_1 - \sigma_2|}{2} = \frac{Y}{2}$$

$$\Rightarrow \sigma_1 - \sigma_2 = \pm Y$$



Ref Fig: Mechanics of Materials, Pytel & Kiusalaas

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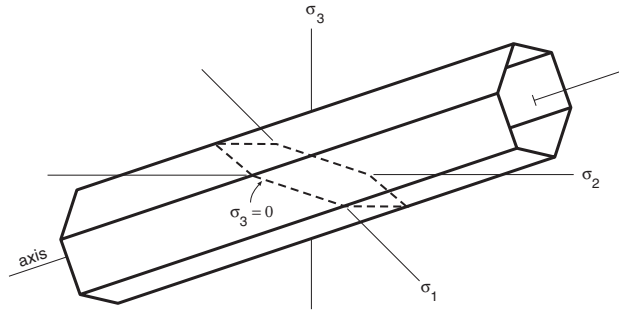
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Yield Criterion

Maximum Shear Stress Criterion (Tresca) :



- In general, $\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} \leq \frac{\sigma_Y}{2} \Rightarrow (\sigma_{\max} - \sigma_{\min}) \leq \sigma_Y$

Ref: Mechanical Behavior of Materials, Dowling

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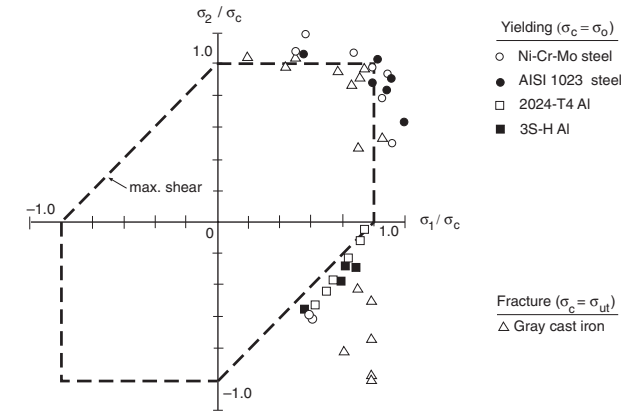
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Maximum Shear Stress Criterion

Experimental Data :



Ref: Mechanical Behavior of Materials, Dowling

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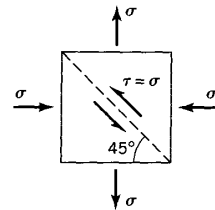
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Maximum Shear Stress Criterion

Tresca Criterion

Drawbacks:

- Very well predicts yield response in many ductile materials along certain stress paths.
- However, under pure shear (torsion test) it over predicts the yield value by about 15%
- The criterion is therefore conservative from a design standpoint.



NOTE: Good for conservative predictions in ductile materials.

Ref: Mechanical Behavior of Materials, Dowling

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Maximum Shear Stress Criterion

[Example 1]: Consider the pipe with closed ends that is subjected to principal stresses $\sigma_1 = 704.3$ MPa, $\sigma_2 = 195.7$ MPa, $\sigma_3 = -20$ MPa. What is the *safety factor* against yielding by *maximum shear stress criterion* if the pipe has a yield strength of 1.791 GPa.

- Factor of Safety:** Ratio of yield strength of the material to effective stress determined based on a yield criterion.
- Based on Maximum shear stress criterion:
 $\sigma_{\text{eff}} = \sigma_{\max} - \sigma_{\min} = 724.3$ MPa
- Since, $\sigma_{\text{eff}} < \sigma_Y$, the pipe has not yielded.
- Factor of safety: $\frac{\sigma_Y}{\sigma_{\text{eff}}} = \frac{1791}{724.3} = 2.47$

Ref: Mechanical Behavior of Materials, Dowling

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Distortional energy density Criterion

von Mises Criterion

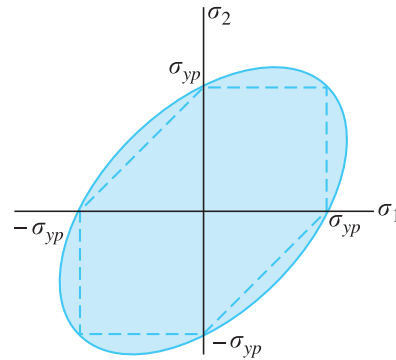
- Yielding occurs when the distortional strain energy density equals that at yield under uniaxial tension, i.e., $U_D = U_{DY}$.
- Strain energy density can be additively decomposed as:

$$U_0 = U_H + U_D$$

where:

$$U_H = \frac{(\sigma_1 + \sigma_2 + \sigma_3)^2}{18K}$$

$$U_D = \frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{12G}$$



Distortional energy density Criterion

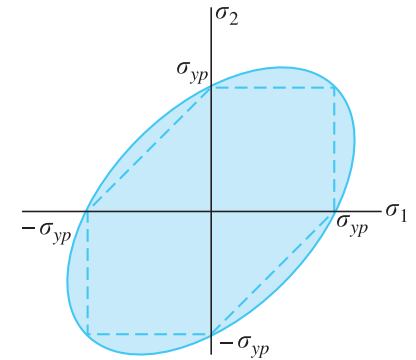
von Mises Criterion

- The distortional strain energy density can also be written as:

$$U_D = \frac{1}{2G} |J_2|$$

- The yield function for the distortional energy therefore can be written as:

$$f = \frac{1}{6} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] - \frac{1}{3} Y^2$$



Ref: Mechanics of Materials, Pytel & Kiusalaas

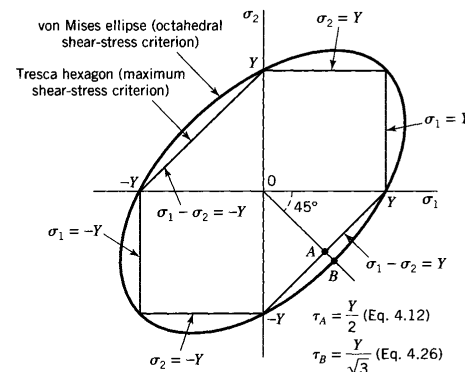
Distortional energy density Criterion

von Mises Criterion

- Defined by the root mean square of the difference between the principal stresses reaching the same value as yielding under simple uniaxial tension.

$$\sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{3}} = \sqrt{\frac{2}{3}} Y$$

$$\sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}} = Y$$



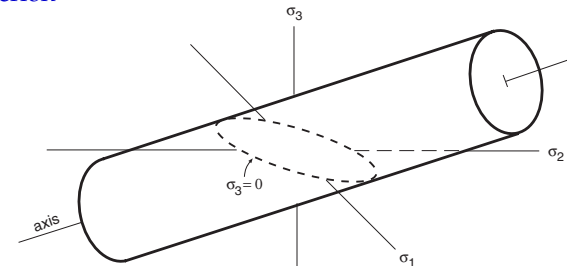
Ref: Advanced Mechanics of Materials, Boreis & Schmidt

Distortional energy density Criterion

von Mises Criterion

- In general Mises criterion in non-principal axes is:

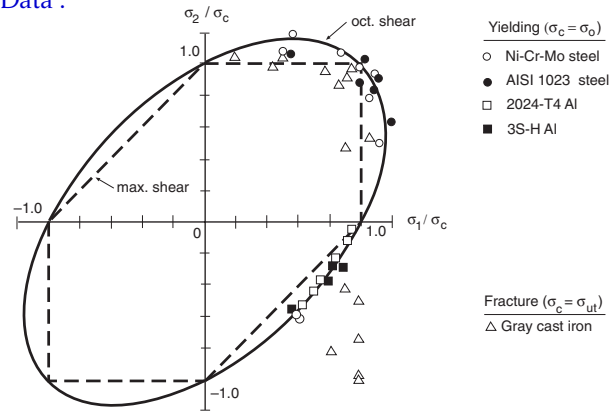
$$\sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2}{2}} + 3(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2) = \sigma_Y$$



Ref: Mechanical Behavior of Materials, Dowling

Distortional energy density Criterion

Experimental Data :



Ref: Mechanical Behavior of Materials, Dowling

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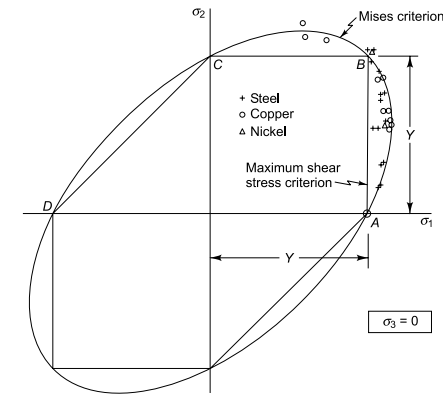
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Distortional energy density Criterion

Experimental Data for Thin-walled tubes : [Figure 5.29]



Ref: An Introduction to Mechanics of Solids, Crandall

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Distortional energy density Criterion

[Example 2] : Consider the pipe with closed ends that is subjected to principal stresses $\sigma_1 = 704.3$ MPa, $\sigma_2 = 195.7$ MPa, $\sigma_3 = -20$ MPa. What is the *safety factor* against yielding by *Mises criterion* if the pipe has a yield strength of 1.791 GPa.

- Based on Mises criterion:

$$\sigma_{\text{eff}} = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$= \sqrt{\frac{(704.3 - 195.7)^2 + (195.7 + 20)^2 + (-20 - 704.3)^2}{2}} = 644.1 \text{ MPa}$$

- Since, $\sigma_{\text{eff}} < \sigma_Y$, the pipe has not yielded.

- Factor of safety: $\frac{\sigma_Y}{\sigma_{\text{eff}}} = \frac{1791}{644.1} = 2.78$

Ref: Mechanical Behavior of Materials, Dowling

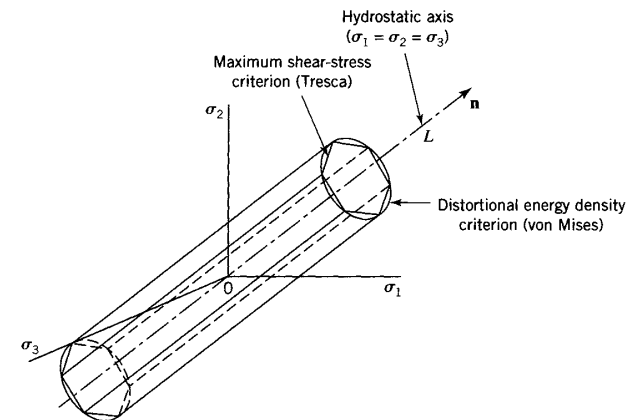
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Tresca & Mises Criterion



Ref: Advanced Mechanics of Materials, Boreis & Schmidt

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Tutorial Problems

[Problem 3.4] A square plate in the side of a ship is subjected to plane strain conditions. If $\sigma_{xx} = \sigma_1 = 500 \text{ MPa}$ and $\epsilon_{xx} = 2\epsilon_{yy}$. Determine the magnitude of $\sigma_{yy} = \sigma_2$ and $\sigma_{zz} = \sigma_3$, assuming linearly elastic conditions. Consider $E = 200 \text{ GPa}$, $\nu = 0.29$.

Answer : $\sigma_{yy} = 377.2 \text{ MPa}$, $\sigma_{zz} = 254.4 \text{ MPa}$

Tutorial Problems

[Example 1] A rectangular plate experiences bi-axial stress state. The thickness of the plate is 0.25 mm . The normal strains measured are: $\epsilon_{xx} = 0.001$ and $\epsilon_{yy} = -0.0007$. If $E = 30 \text{ MPa}$ and $\nu = 0.3$ then determine the stresses and change in thickness of the plate.

Answer :

$$\sigma_{xx} = 26.043 \text{ KPa}; \quad \sigma_{yy} = -13.186 \text{ KPa}; \quad \epsilon_{zz} = -128.5 \times 10^{-6}; \quad \Delta t = -32.1 \times 10^{-6} \text{ mm}$$

Tutorial Problems

[Example 2] For an element in plane stress, normal strains are ϵ_{xx} , ϵ_{yy} and ϵ_{zz} . Then:

- ➊ Obtain ϵ_{zz} as $f(\epsilon_{xx}, \epsilon_{yy})$.
- ➋ Obtain the formula for dilation as $f(\epsilon_{xx}, \epsilon_{yy}, \nu)$.

Answer :

$$\epsilon_{zz} = -\frac{\nu}{(1-\nu)}(\epsilon_{xx} + \epsilon_{yy}); \quad e = \frac{(1-2\nu)}{(1-\nu)}(\epsilon_{xx} + \epsilon_{yy})$$

Tutorial Problems

[Example 3] For a rectangular plate, the following values are given: $\sigma_{xx} = 24 \text{ MPa}$, $\sigma_{yy} = 12 \text{ MPa}$, $\epsilon_{xx} = 440 \times 10^{-6}$ and $\epsilon_{yy} = 80 \times 10^{-6}$. Determine E and ν .

Answer :

$$\nu = 0.35; \quad E = 45 \text{ GPa}$$

Tutorial Problems

[Example 4] A steel rectangular plate $400 \times 800 \times 200$ mm is subjected to normal stress $\sigma_{xx} = 90$ MPa, $\sigma_{yy} = -20$ MPa. Consider $E = 200$ GPa and $\nu = 0.3$. Determine;
a) Max. in-plane shear strain $\gamma_{\max}$ in the plate.

Answer :

$$\gamma_{\max} = 715 \times 10^{-6}$$

Tutorial Problems

[Example 4] b) change in thickness

Answer :

$$\Delta t = -21 \times 10^{-4} \text{ mm}$$

Tutorial Problems

[Example 4] c) change in volume

Answer :

$$\Delta V = 8960 \text{ mm}^3$$

Tutorial Problems

[Example 5] A brass cube with edge length 50 mm is compressed by $P = 175$ kN in two orthogonal directions. Find change in volume and strain energy.

Answer :

$$\Delta V = -56 \text{ mm}^3; \quad U = 4.04 \text{ J}$$

Reading Assignment

Chapter 4: Section 4.1 - 4.3